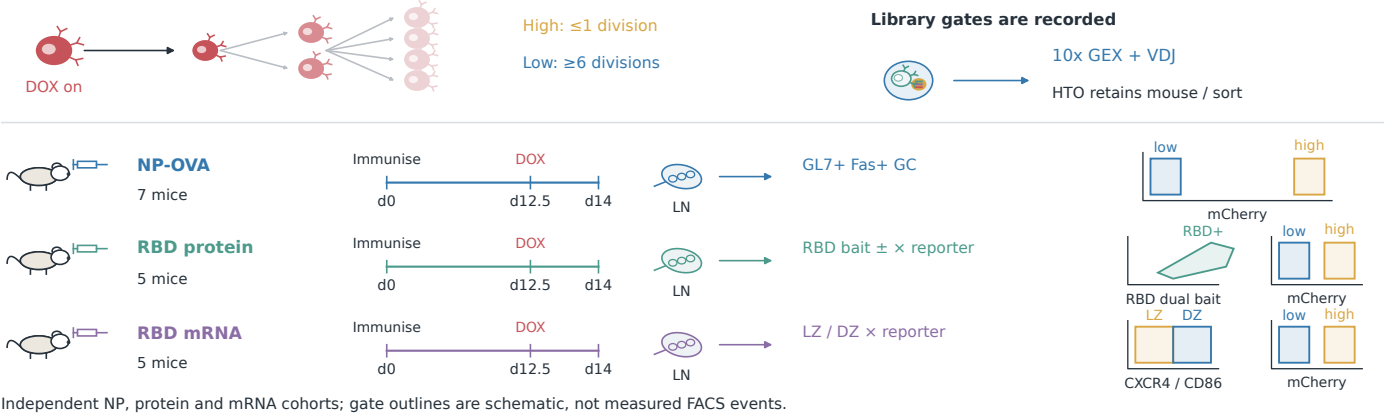


Figure 2 | Model-antigen GC state nodes and independent reporter measurements

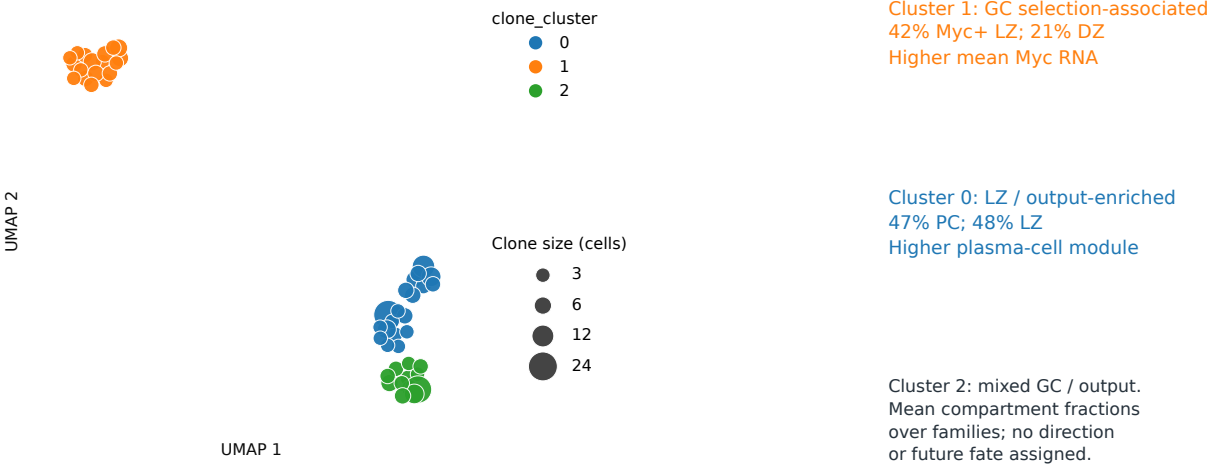
Clone embedding is coloured by unsupervised Leiden clusters; measured gates and RNA provide GC-state interpretation.

A Independent NP-OVA and RBD reporter designs

H2B-mCherry reporter: stop synthesis, read 36 h of division history



B NP-OVA day 14: clone embedding and Leiden reclustering



GSE246382: 49 same-mouse sequence-defined families / 260 cells; ≥3 cells per family; area follows capture count.

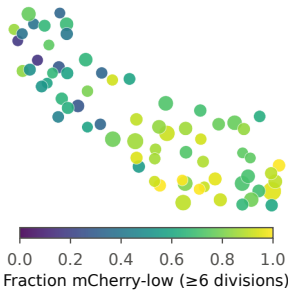
Separate reporter evidence

NP-OVA division reporter
36-hour H2B-mCherry window

C: measured division bias
D: receptor mutation history

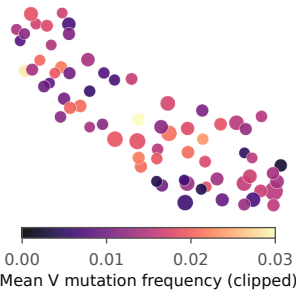
Independent of the B cohort.

C NP-OVA: division-associated clone-state bias



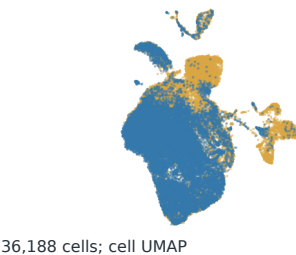
69 reliable clones; clone UMAP

D NP-OVA: mutation history on the same clone map



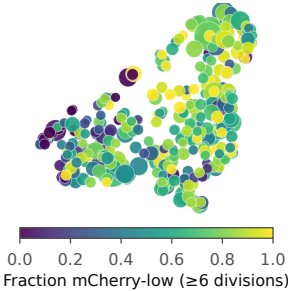
69 reliable clones; clone UMAP

E RBD: cell division gate



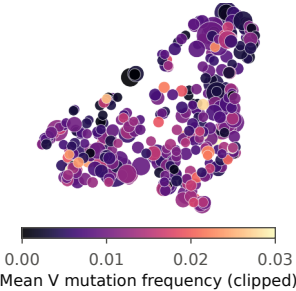
36,188 cells; cell UMAP

F RBD: division-associated clone-state bias



381 reliable clones; clone UMAP

G RBD: mutation history on the same clone map



381 reliable clones; clone UMAP